Lesson 01: Methods of Knowing and Understanding Science

English Stage 4 (Psychology) · Unit 1: Psychology as a Science

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 1.1 Prelude to the Science of Psychology; 1.2 Methods of Knowing; 1.3 Understanding Science.

In this lesson

You will compare five ways of acquiring knowledge, learn the three features that make psychology a science, and see how science differs from pseudoscience. On the language side, you will practice defining terms the way the textbook does and expressing cautious conclusions with hedges.

Learning goals

After this lesson you should be able to:

- Describe the five methods of knowing, with one strength and one problem of each.
- Explain the three fundamental features of science and why psychology is a science.
- Decide whether a claim is falsifiable and explain why a belief is pseudoscientific.
- Write a one-sentence academic definition of a research term.
- Recognize and use hedges in conclusions.

Key language

Term or phrase	Meaning	Example
intuition	relying on our guts, emotions or instincts; believing what feels true	Your friend will not look at you, so you decide she is lying.
authority	accepting ideas because an authority figure (parent, doctor, professor, the media) says they are true	Your parents told you to make your bed.
rationalism	using logic: stating premises and following logical rules to reach conclusions	All swans are white; this is a swan; so it is white.
empiricism	acquiring knowledge through observation and experience	The world looks flat, so people believed it was flat.
scientific method	a process of systematically collecting and evaluating evidence to test ideas and answer questions	counting the words that 369 students actually spoke
science	a general way of understanding the natural world featuring systematic empiricism, empirical questions and public knowledge	Psychology takes this approach to behavior.

Term or phrase	Meaning	Example
systematic empiricism	learning about the world through careful observation	planning, making, recording and analyzing observations
empirical question	a question about the way the world actually is that can be answered by making systematic observations	Do women talk more than men?
public knowledge	detailed descriptions of research that are available to others, usually through publication in a professional journal	Mehl's study appeared in *Science*.
replication	conducting a study again, exactly or with modifications, to see if the same results are observed	The handwashing effect did not appear again.
pseudoscience	beliefs or activities claimed to be scientific that lack one or more of the three features of science	biorhythms, astrology
falsifiable	a claim is falsifiable if there is an observation that would, if it were made, count as evidence against it	"Women talk more than men" is falsifiable.
on the basis of available evidence	judging from the evidence we have now	"We therefore conclude, on the basis of available empirical evidence, that ..."
To be clear, ...	stops the reader from drawing a conclusion that is too strong	"To be clear, at this stage we are still unable to definitively conclude ..."

Rules and explanations

The ideas

Why psychology needs science (1.1). A widely cited estimate says that women speak 20,000 words a day and men 7,000. Mehl and colleagues found that no one had actually counted. In their study, 369 college students wore audio recorders: women spoke an average of 16,215 words a day and men 15,669, an extremely small difference that could easily be explained by chance. They concluded that the stereotype is "unfounded".

Five methods of knowing (1.2)

Method	Main problem	But ...
Intuition	driven by cognitive and motivational biases, not logic or evidence	intuitive decisions are sometimes better than analysis (Gladwell's *Blink*)
Authority	authorities may be wrong, may rely on intuition, or may want to mislead you (a made bed is a warm, damp home for mites)	we cannot check everything ourselves; we can evaluate their credentials, methods and motives

Method	Main problem	But ...
Rationalism	wrong premises or errors in logic give invalid conclusions (there are black swans in Australia)	correct premises and logic make it a sound method
Empiricism	our experience is limited, our senses deceive us, and prior experience changes what we perceive	it is at the heart of the scientific method
Scientific method	not always feasible (time, resources); answers only empirical questions	the most likely method to produce valid knowledge

Scientists may use the first four to generate ideas, but they test them with systematic empiricism (careful observation under controlled conditions) and use rationalism to reach valid conclusions.

Three features of science (1.3; Stanovich, 2010). The sciences share neither subject matter nor equipment; they share a general approach to understanding the natural world.

1. **Systematic empiricism.** Scientists check their ideas against planned, recorded observations. When Mehl's data conflicted with the stereotype, the researchers trusted the data.
2. **Empirical questions.** "Is this stereotype accurate?" can be answered by observation. "Is it wrong to hold inaccurate stereotypes?" is a value judgment about how the world *ought* to be, which science cannot answer.
3. **Public knowledge.** Scientists publish their methods and results, increasingly in open access journals that anyone can read. Publication matters because science is a social process (a collaboration across time and space) and because it makes science **self-correcting**.

Self-correction in action. Schnall and colleagues (2008) reported that washing one's hands leads people to judge moral transgressions as less wrong. With the same materials, nearly identical procedures and a much larger sample, Johnson, Cheung and Donnellan did not find the effect, suggesting that the original result may have come from a small sample. Still, we cannot definitively conclude that the effect does not exist.

> **Accuracy note:** The book presents this replication as part of the "Many Labs" project, which tested 13 effects. In fact, the handwashing effect was not one of those 13: Johnson, Cheung and Donnellan's study was a separate replication, published in the same journal issue as Many Labs. The point about self-correction is unchanged.

Science versus pseudoscience. A belief is pseudoscientific if (a) its supporters claim or imply that it is scientific but (b) it lacks one or more features of science:

- no systematic empiricism: there is no research, or it is ignored (biorhythms, supposed 23-, 33- and 28-day cycles, remain popular although the evidence indicates they do not exist);
- no public knowledge: research is claimed but never published for others to evaluate;
- no empirical questions: the claims are not **falsifiable** (Karl Popper). If psychic powers "disappear when observed too closely", success and failure in a test are both consistent with the claim.

Pseudoscience matters to psychology because many pseudosciences (astrology, graphology, magnet therapy) claim to explain behavior, and because such beliefs can do real harm (homeopathy instead of an empirically supported treatment).

Language focus

1. Defining a term. A good definition names a general class (a process, an attitude, a question) and then what makes the term different. The book's patterns:

- *X involves + -ing*: "Rationalism **involves using** logic and reasoning to acquire new knowledge."
- *X refers to + noun*: "Pseudoscience **refers to** activities and beliefs that are claimed to be scientific ... but are not."
- *X is a + class noun*: "The scientific method **is a process of** systematically collecting and evaluating evidence ..."

2. Hedging a conclusion. Careful writers show exactly how strong their evidence is.

- "We therefore conclude, **on the basis of available empirical evidence**, that ... the stereotype ... is unfounded."
- "..., **suggesting that** the original finding **may have stemmed from** the relatively small sample size"
- "To be clear, **at this stage we are still unable to definitively conclude** that the handwashing effect does not exist."

Hedges include modal verbs (*may, might, can*), verbs such as *suggest*, adverbs such as *allegedly*, and limits such as *at this stage*.

> **Note for Arabic speakers** - General academic subjects take no article: "Psychology is a science", "Science relies on observations" (not "The psychology is the science"). Use *the* for something specific: "the scientific method". - After *involve*, use the -ing form: "Empiricism involves acquiring knowledge" (not "involves to acquire").

Practice

1. Reading questions

1. What did Mehl and colleagues find, and why did they call the stereotype "unfounded"? (1.1)
2. Why do we still acquire so much knowledge through authority, and what can we evaluate about an authority figure? (1.2)
3. What two drawbacks of the scientific method does Section 1.2 name?
4. What do astronomy, biology and chemistry have in common, if not their subject matter or equipment? (1.3)
5. Give the two reasons why publication is an essential feature of science.
6. Why is the theory of biorhythms pseudoscientific?
7. Why is the claim that psychic powers disappear when observed too closely a problem for science?

8. Give the three reasons for concerning ourselves with pseudoscience.

2. Book task. Sections 1.1 to 1.3 have no exercises (Chapter 1's exercises are in Lesson 03). Do the task that opens Section 1.2: "Take a minute to ponder some of what you know and how you acquired that knowledge." Start with the book's three examples (you should make your bed; swans are white; your friend is lying): which method produced each belief, and should we trust it?

3. Language exercise

A. Turn each description into a one-sentence definition with the pattern given.

1. Science relies on structured observations. (systematic empiricism; *refers to*)
2. When we use intuition, we rely on our guts, emotions and instincts. (intuition; *involves*)
3. Researchers conduct a study again to see if the same results are observed. (replication; *is the process of*)

B. Find the hedge in each sentence from 1.3 and say what it does.

1. "If reliable, this effect might help explain why so many religious traditions associate physical cleanliness with moral purity."
2. "Allegedly, the physical cycle has a period of 23 days."

C. Rewrite with at least two hedges: "The replication proves that the handwashing effect does not exist."

Speak with your tutor

- "Here is my definition of systematic empiricism: [definition]. Correct it."
- "Give me five claims about behavior, one at a time. I will say whether each is an empirical question and whether it is falsifiable, and why."
- "Pretend you believe in biorhythms. I will use the three features of science to explain politely why the theory is pseudoscience."
- "In PSYC 110 Lecture 3 (Foundations: Freud), Professor Bloom uses Popper's idea of falsifiability to say that astrological predictions 'can't be wrong' and that much of Freudian theory is 'not even wrong'. Help me compare his examples with the ESP example in Section 1.3."

Writing task

Project warm-up. Your research project (a proposal on sleep and memory in university students, or a topic of your own) starts in Lesson 04. Write one paragraph (150 to 200 words) about what you already believe about your topic and how you know it.

Include:

- two or three beliefs about the topic (for example, "Students remember less after a night without sleep");
- for each belief, the method of knowing behind it and one problem with that method;
- whether each belief is an empirical question and whether it is falsifiable;

- one definition of a key term, using a pattern from the Language focus;
- a final hedged sentence about what you would need to find out.

Self-editing checklist:

- Did I name a method of knowing for each belief, with one problem?
- Did I say whether each belief can be tested by systematic observation?
- Does my definition use a class noun, *involves* or *refers to*?
- Did I use *involve* + -ing and no article before general subjects ("memory", "sleep")?
- Is my final sentence hedged?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- "The Replication Crisis in Psychology", a Noba module linked in Section 1.3 (reading): https://nobaproject.com/modules/the-replication-crisis-in-psychology
- *The Skeptic's Dictionary*: https://www.skepdic.com
- PSYC 110 Lecture 3, Foundations: Freud (falsifiability and astrology in Chapter 6, from 32:56): https://oyc.yale.edu/psychology/psyc-110/lecture-3
- Note: the book's link to *Neurobonkers* now leads to an unrelated website; do not use it.

Answer key

1. Reading questions

1. Women spoke an average of 16,215 words a day and men 15,669, an extremely small difference that could easily be explained by chance. Their systematic observations did not support the claimed 20,000 versus 7,000 (1.1).
2. We do not have time to question and research everything ourselves. We can evaluate their credentials, the methods they used to reach their conclusions, and whether they have reasons to mislead us (1.2, "Authority").
3. It is not always feasible, because it can require considerable time and resources, and it can answer only empirical questions (1.2, "The Scientific Method").
4. A general approach to understanding the natural world (1.3, "What Is Science?").
5. Science is a social process, a large-scale collaboration among researchers across time and space; and publication makes science self-correcting, because others can detect and correct errors (1.3).
6. Its adherents present it as scientific, with scientific-sounding terms such as *sinusoidal wave*, but it lacks systematic empiricism: the relevant research indicates that biorhythms do not exist, and it is ignored (1.3, "Science Versus Pseudoscience").
7. It makes the claim unfalsifiable: success would support it, and failure would also be "consistent" with it, so no possible observation could count against it (1.3).
8. It brings the features of science into sharper focus; pseudoscientific beliefs are widely held and can cause great personal harm, so we need to identify and evaluate them; and many claim to explain behavior, so psychology must distinguish itself from them (1.3).

2. Book task. Making your bed: authority (your parents); here it is wrong, because a made bed gives mites the warm, damp environment in which they thrive. White swans: empiricism (the swans you have seen); wrong, because there are black swans in Australia. A lying friend: intuition; she may simply be holding in gas or be preoccupied with something else (1.2). Your own examples will vary.

3. Language exercise

A. (example answers)

1. Systematic empiricism refers to learning about the world through structured, careful observation.
2. Intuition involves relying on our guts, emotions and instincts and believing what feels true.
3. Replication is the process of conducting a study again, exactly or with modifications, to see if the same results are observed.

B. 1. "If reliable" (a condition: the effect is not established) and "might" (a possibility, not a fact). 2. "Allegedly" (the author reports what believers claim without accepting it).

C. (example answer) "The replication suggests that the handwashing effect may not be reliable, but at this stage we cannot conclude that it does not exist."

My notes